

ASSIGNMENT

24MCAT142 - Advanced Web Programming and
Digital Public Infrastructure

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Roll No : 46

1. Shared Hosting vs VPS vs Cloud Hosting

Scenario: A small business website is facing slow performance as traffic increases.

Activity: Compare shared, VPS, and cloud hosting in terms of cost, performance, and scalability. Recommend the best solution.

Example Reference: (Shared Hosting), (VPS), (Cloud Hosting)

Main Issues:

- Website takes longer to load
- Sometimes crashes during high traffic
- Users may leave due to slow experience

Root cause: The current hosting cannot handle increased traffic efficiently.

Analysis of Hosting Types

1. Shared Hosting

- Multiple websites share one server
- Cost: Very low
- Performance: Slow when traffic increases
- Scalability: Poor

Problem: Resources are shared → performance drops when others use more

2. VPS (Virtual Private Server)

- Server divided into virtual parts
- Cost: Medium
- Performance: Better than shared
- Scalability: Limited

Improvement: Dedicated portion of resources → more stable performance

3. Cloud Hosting

- Uses multiple connected servers
- **Cost:** Pay as you use
- **Performance:** High
- **Scalability:** Excellent

Advantage: Can handle sudden traffic spikes easily

The best solution is Cloud Hosting

Because,

- ❓ Handles high traffic easily
- ❓ Auto-scales resources
- ❓ Reduces downtime
- ❓ Better user experience

2. E-commerce Website During Peak Traffic

Scenario: An online store faces heavy traffic during a festival sale and risks downtime.

Activity: Design a hosting architecture to handle peak load with scaling and load balancing.

Example Reference: handles millions of users during sale events

During a festival sale, the e-commerce website gets **very high traffic**. This can cause:

- Slow loading
- Server crashes (downtime)
- Loss of customers and sales

The main issue is the system **cannot handle sudden peak load**.

Use a **cloud-based scalable architecture**:

- **Load Balancer** → Distributes users across multiple servers
- **Multiple Servers** → Handles requests instead of one server
- **Auto Scaling** → Adds more servers when traffic increases
- **Cloud Hosting** → Ensures high availability
- availability

◆ How It Improves Performance

- Prevents server overload
- Handles millions of users smoothly
- Reduces downtime
- Keeps website fast during peak sales

3. Website Migration (Shared to Cloud)

Scenario: A company plans to migrate its website to cloud hosting for better performance.

Activity: Prepare a migration plan including backup, testing, and downtime minimization.

Example Reference: migrated to cloud infrastructure for scalability

The company is using shared hosting, which causes:

- Slow performance
- Limited scalability
- Difficulty handling more users

Goal: Move to cloud hosting for **better speed and scalability**.

Migration Plan

- **Backup Data** → Take full backup of website files and database
- **Set Up Cloud Server** → Configure environment in cloud
- **Migrate Data** → Upload files and restore database
- **Testing** → Check website functionality and performance
- **Switch to Cloud** → Update domain to point to new server

◆ Downtime Minimization

- Perform migration during low-traffic time
- Test before going live
- Use temporary parallel setup (old + new server)

◆ Benefits After Migration

- Faster website
- Better scalability
- Improved reliability

Migrating to cloud hosting improves performance and ensures the website can handle future growth with minimal downtime.

4. Cost Optimization in Hosting

Scenario: A startup is experiencing high cloud bills.

Activity: Suggest a cost-effective hosting plan with optimized resource usage.

Example Reference: optimizes cloud costs using scalable resources

The startup is using cloud hosting, but:

- Monthly bills are very high
- Resources may be unused or over-allocated

Main issue: **Inefficient resource usage leading to high cost**

◆ Solution (Cost-Effective Plan)

- **Auto Scaling** → Use resources only when needed
- **Right-Sizing** → Choose correct server size (avoid overuse)
- **Serverless Options** → Pay only for actual usage
- **Monitoring Tools** → Track and reduce unnecessary usage

◆ How It Reduces Cost

- Avoids paying for idle servers
- Adjusts resources based on traffic
- Improves overall efficiency

By using **auto scaling and proper resource management**, the startup can reduce cloud costs while maintaining performance.

5. Managed vs Unmanaged Hosting

Scenario: A startup lacks technical expertise to manage servers.

Activity: Conduct a role play to decide the best hosting option.

Example Reference: provides managed hosting solutions

◆ Understanding the Problem

The startup:

- Does not have technical experts
- Cannot manage servers, security, or updates

Main issue: **Lack of technical knowledge to handle hosting**

◆ Role Play (Decision Discussion)

Startup Owner:

“We need a website, but we don’t know how to manage servers.”

Technical Advisor:

“You have two options: Managed and Unmanaged hosting.”

Owner:

“What’s the difference?”

Advisor:

- **Managed Hosting:** Provider handles setup, updates, security
- **Unmanaged Hosting:** You must manage everything yourself

Owner:

“Then managed hosting is better for us.”

◆ Comparison

- **Managed Hosting:** Easy to use, less responsibility, slightly higher cost
- **Unmanaged Hosting:** Full control, requires technical skills, lower cost

◆ Recommended Solution

Managed Hosting

Why?

- No need for technical expertise
- Provider handles maintenance and security
- Saves time and reduces risk

For startups without technical knowledge, **managed hosting is the best choice** as it simplifies operations and ensures reliability.

6. Cloud Hosting Architecture Design

Scenario: A company wants to launch a scalable global web application.

Activity: Draw and explain a cloud architecture (load balancer, database, CDN).

Example Reference: uses distributed cloud architecture on.

◆ Understanding the Problem

The company wants to launch a **global web application** that:

- Works fast for users worldwide
- Handles high traffic
- Remains available all the time

Main need: **Scalable and globally distributed system**

Solution (Cloud Architecture)

- **Load Balancer** → Distributes traffic across servers
- **Application Servers** → Handle user requests
- **Database (Cloud DB)** → Stores data reliably
- **CDN (Content Delivery Network)** → Delivers content faster worldwide

◆ How It Works

- User request goes to nearest CDN server
- Load balancer sends request to available server
- Server processes request and fetches data from database
- Response is delivered quickly to user

◆ Benefits

- Fast performance globally
- High scalability
- High availability (less downtime)

Using a **distributed cloud architecture with CDN and load balancing** ensures the application is fast, scalable, and reliable worldwide.

7. CDN and Website Performance

Scenario: A website is slow for international users.

Activity: Analyze and propose a CDN-based solution.

Example Reference: improves global website speed

◆ Understanding the Problem

The website works fine locally but is **slow for international users** because:

- Data travels long distance from the main server
- Higher latency causes delay in loading

Main issue: **No global content distribution**

◆ Solution (CDN-Based Approach)

Use a **Content Delivery Network (CDN)**:

- Store website content on multiple **edge servers worldwide**
- Deliver data from the **nearest server to the user**
- Cache static content (images, CSS, JS)

◆ How It Improves Performance

- Reduces latency (faster loading)
- Improves user experience globally
- Reduces load on main server

Implementing a **CDN** ensures fast and reliable website access for users across different countries.

8. Security in Web Hosting

Scenario: A website suffers a data breach due to weak security.

Activity: Conduct a security audit and recommend improvements.

Example Reference: data breach case highlights security risks

◆ Understanding the Problem

The website experienced a **data breach** due to weak security:

- Sensitive data exposed
- Poor protection against attacks
- Possible loss of user trust

Main issue: **Lack of proper security measures**

◆ Security Audit & Solution

- **Enable SSL (HTTPS)** → Encrypt data between user and server
- **Use Firewall & WAF** → Block malicious traffic
- **Regular Updates** → Keep software and plugins updated
- **Strong Authentication** → Use strong passwords & 2FA
- **Backup System** → Regular data backups

◆ How It Improves Security

- Protects user data
- Prevents unauthorized access
- Reduces risk of future attacks

A proper security audit with strong protection measures ensures the website is safe and reliable.

9. SaaS Application Hosting

Scenario: A team develops an online exam system for multiple users.

Activity: Design a SaaS hosting model ensuring scalability and availability.

Example Reference: operates as a SaaS platform

◆ Understanding the Problem

The team built an online exam system used by many users at the same time:

- High number of concurrent users
- Needs continuous availability (no downtime)
- Must scale during exam time

Main issue: Handling multiple users efficiently with high availability

◆ **Solution (SaaS Hosting Model)**

- Multi-Tenant Architecture → Multiple users share one application securely
- Load Balancer → Distributes traffic across servers
- Auto Scaling → Adds servers during peak exam time
- Cloud Database → Stores user data reliably
- Backup & Monitoring → Ensures uptime and quick recovery

◆ **How It Improves System**

- Supports many users at once
- Ensures high availability
- Scales automatically during exams

A **cloud-based SaaS model with multi-tenancy and auto scaling** ensures the exam system is scalable, reliable, and always available.

10. Website Failure (Downtime Case)

Scenario: A major platform experiences global outage due to configuration issues.

Activity: Perform root cause analysis and suggest preventive strategies.

Example Reference: outage due to DNS misconfiguration.

◆ **Understanding the Problem**

A major platform faced a **global outage** due to configuration issues (like DNS error):

- Website not accessible worldwide
- Loss of users and revenue
- Damage to reputation

Main issue: **Incorrect configuration without proper validation**

◆ **Root Cause Analysis**

- **DNS Misconfiguration** → Wrong routing of domain requests
- **No Testing Before Deployment** → Changes applied directly
- **Lack of Monitoring** → Issue not detected quickly

◆ **Preventive Strategies**

- **Configuration Testing** → Test changes before deployment

- **Backup & Rollback Plan** → Quickly revert if failure occurs
- **Monitoring & Alerts** → Detect issues instantly
- **Redundant Systems** → Use multiple DNS/servers for reliability

Proper testing, monitoring, and backup strategies can prevent outages and ensure high availability.

11. Enterprise/Government Cloud Adoption

Scenario: A government system plans to migrate to cloud infrastructure.

Activity: Debate advantages and challenges of cloud adoption.

Example Reference: cloud adoption initiatives

◆ Understanding the Problem

A government system plans to move from traditional infrastructure to cloud:

- Needs better scalability and performance
- Must ensure **data security and compliance**
- Requires reliable and continuous service

Main concern: **Balancing benefits of cloud with security and control**

◆ Advantages vs Challenges

Advantages

- **Scalability** → Easily handle large public systems
- **Cost Efficiency** → No need for heavy infrastructure
- **Accessibility** → Services available anytime, anywhere
- **Disaster Recovery** → Better backup and recovery options

Challenges

- **Data Security & Privacy** → Sensitive government data risk
- **Compliance Issues** → Must follow strict regulations
- **Dependency on Provider** → Less direct control
- **Migration Complexity** → Time and planning required

◆ Recommended Approach

Use a **Hybrid Cloud Model**:

- Keep sensitive data on private servers
- Use public cloud for scalable services

Cloud adoption offers scalability and efficiency, but governments must ensure strong security and compliance using a hybrid approach.

12. Local Business Website Hosting

Scenario: A small business wants to launch a website with minimal cost.

Activity: Recommend a suitable hosting solution with justification.

Example Reference: websites hosted on affordable shared hosting

◆ Understanding the Problem

A small business wants to launch a website:

- Has **low budget**
- Expects **limited traffic** initially
- Needs simple and easy setup
- Main goal: **Affordable and basic hosting solution**

◆ Recommended Solution

Shared Hosting

◆ Justification

- **Low Cost** → Best for small budget
- **Easy to Use** → No technical knowledge required
- **Sufficient for Low Traffic** → Works well for small websites

◆ Future Scope

- Upgrade to VPS or cloud when traffic increases
- Add performance improvements later

Shared hosting is the best starting option for small businesses due to its low cost and simplicity.